



Meta-Analyses for Movement

Continuous-Time Animal Movement Modeling

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Overview

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- 1) Hierarchical vs. non-hierarchical models
- 2) Individual to population-level inferences
- 3) Applications for animal movement meta-analyses

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- 1) **Hierarchical vs. non-hierarchical models**
- 2) Individual to population-level inferences
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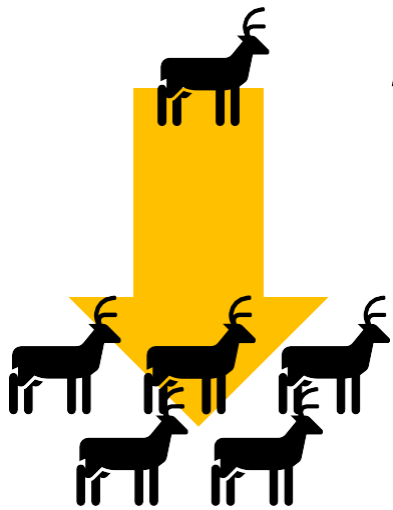
Hierarchical vs non-hierarchical

- Analyses of ecological data should always account for the *uncertainty* in the process(es) that generated the data.

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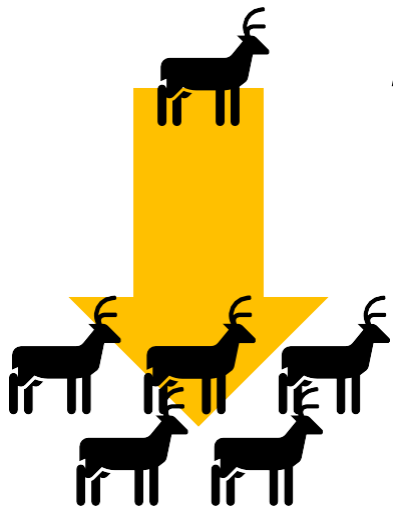
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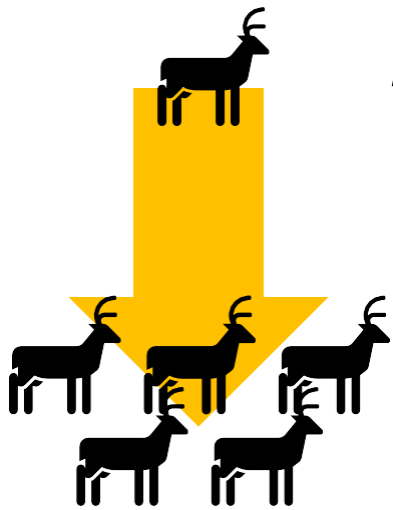
We want to quantify the effect of covariates, such as *species*, *sex*, *body size*, *age*, *habitat*, *anthropogenic impact*, etc....



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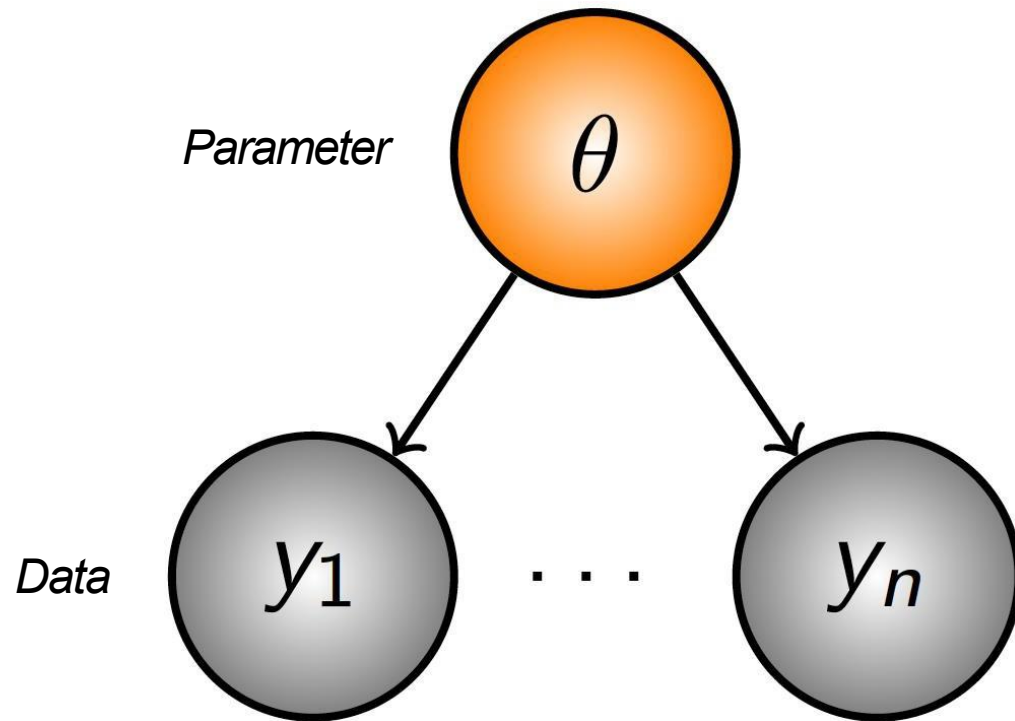
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We want to quantify the effect of covariates, such as *species, sex, body size, age, habitat, anthropogenic impact*, etc....

...even if we are comparing different populations with different **movement behaviors** or **sampling schedules**.

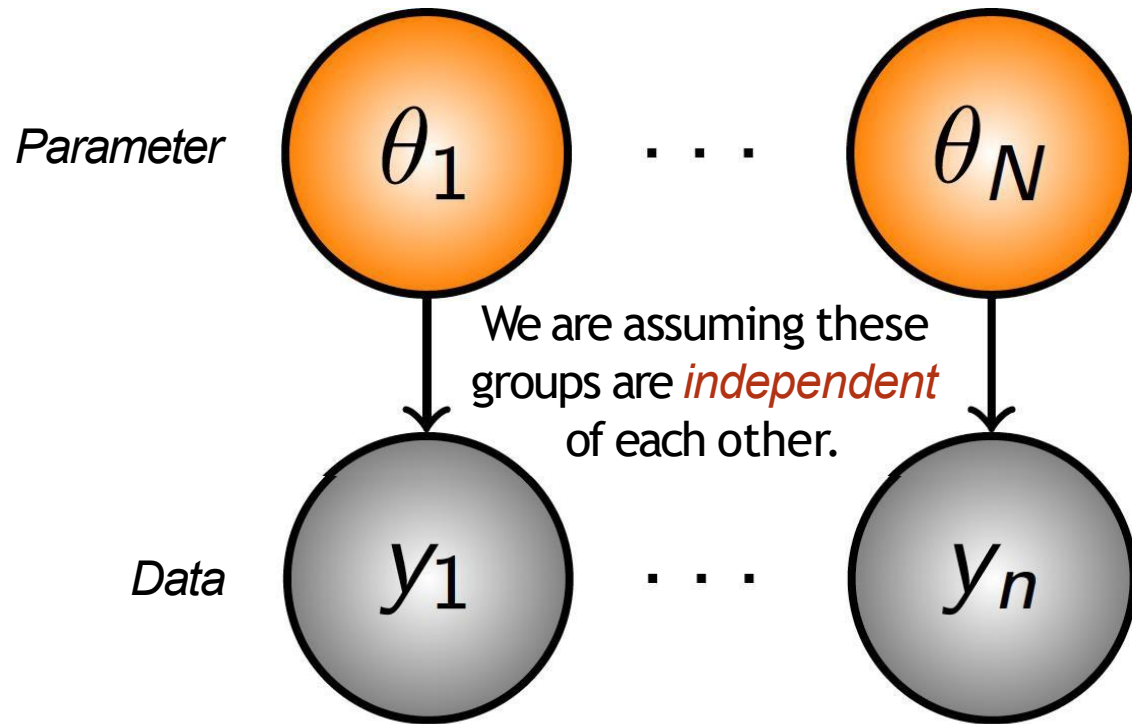
Hierarchical vs non-hierarchical



Non-hierarchical models

- How does data inform parameters?
 - Complete **pooling**

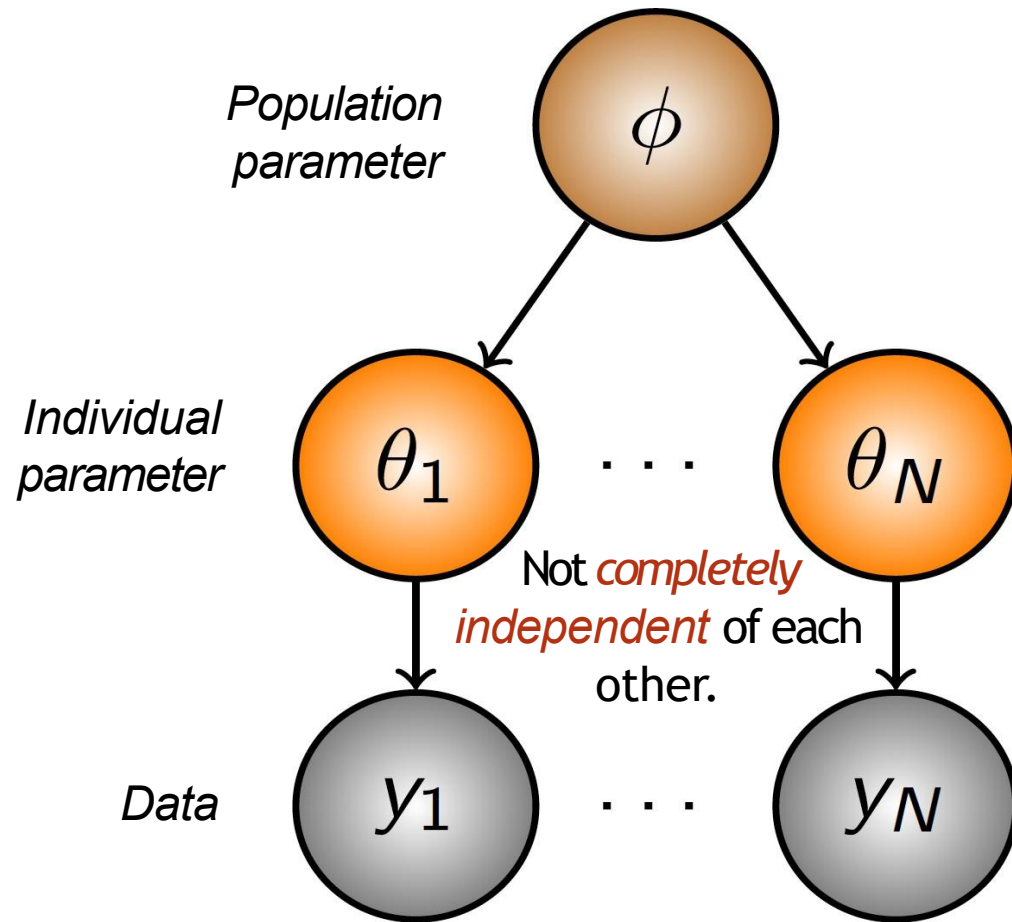
Hierarchical vs non-hierarchical



Non-hierarchical models

- How does data inform parameters?
 - Complete **pooling**
 - No **pooling** (no sharing of information)

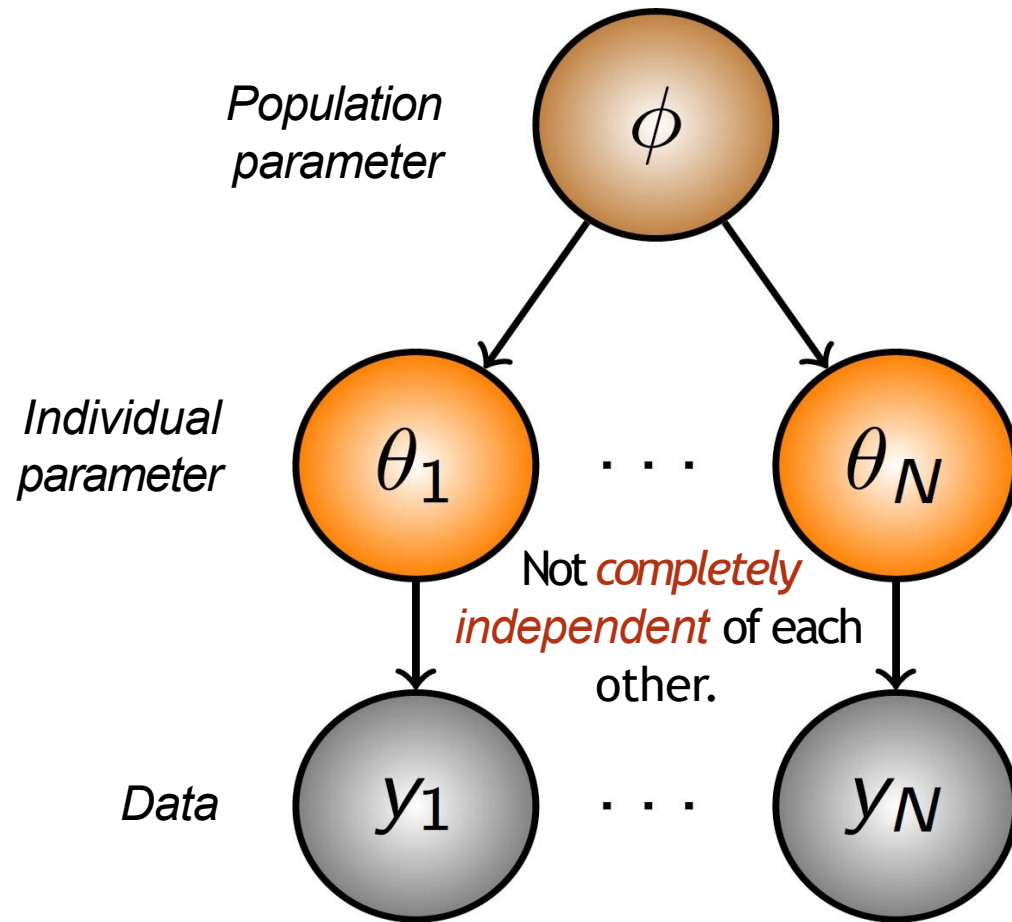
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Hierarchical models

- How does data inform parameters?
 - **Parameters are not independent** of each other

Hierarchical vs non-hierarchical



Hierarchical models

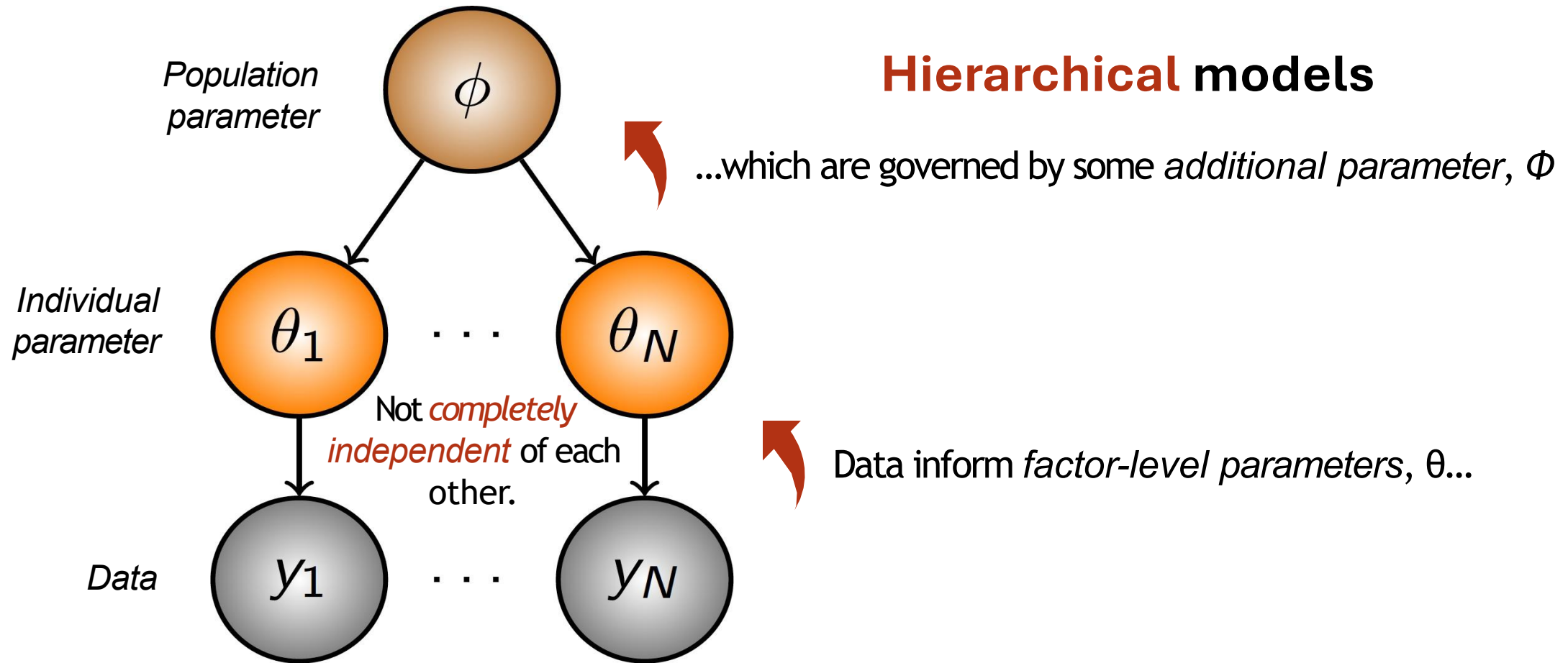
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Data inform *factor-level parameters*, $\theta \dots$

Hierarchical vs non-hierarchical

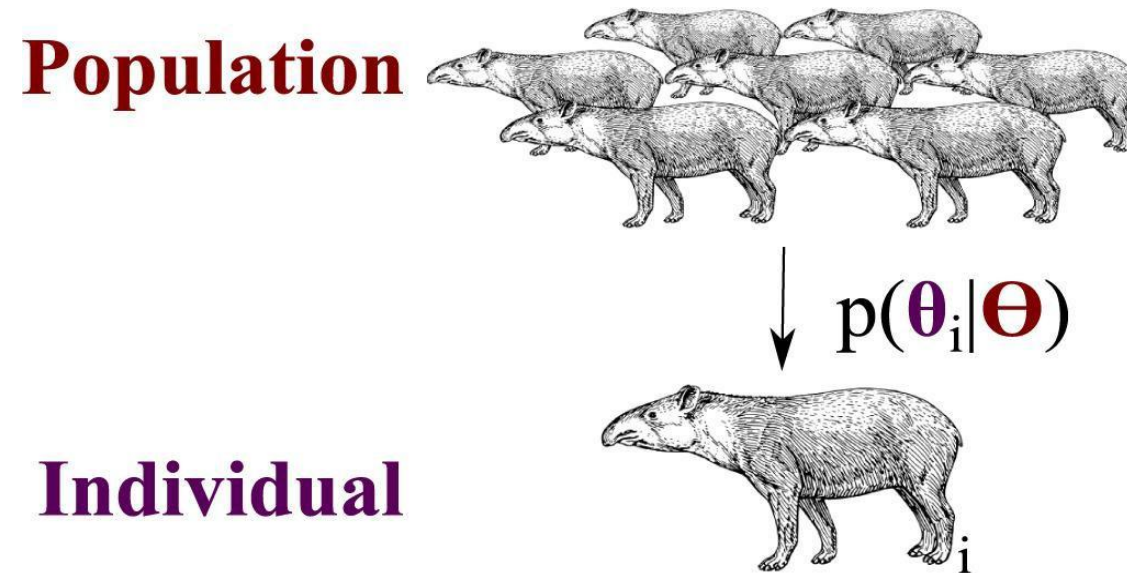
Hierarchical models



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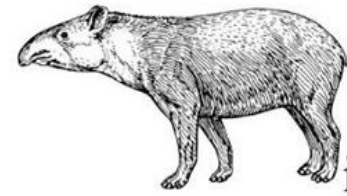
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Individual to population-level inferences



Individual to population-level inferences

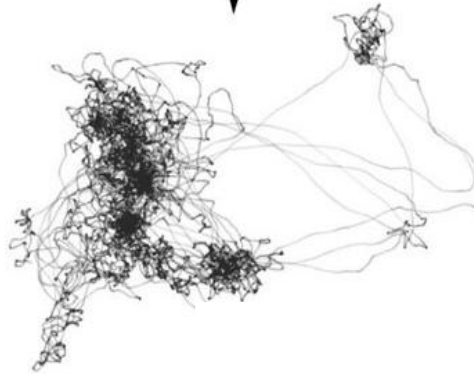
Individual



$$\downarrow p(\theta_i | \Theta)$$

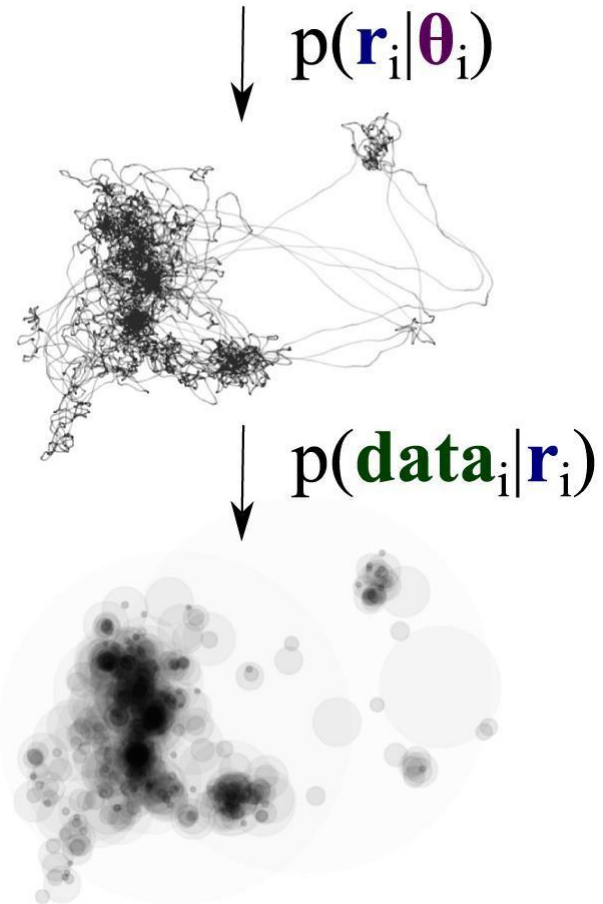
$$\downarrow p(\mathbf{r}_i | \theta_i)$$

Trajectory



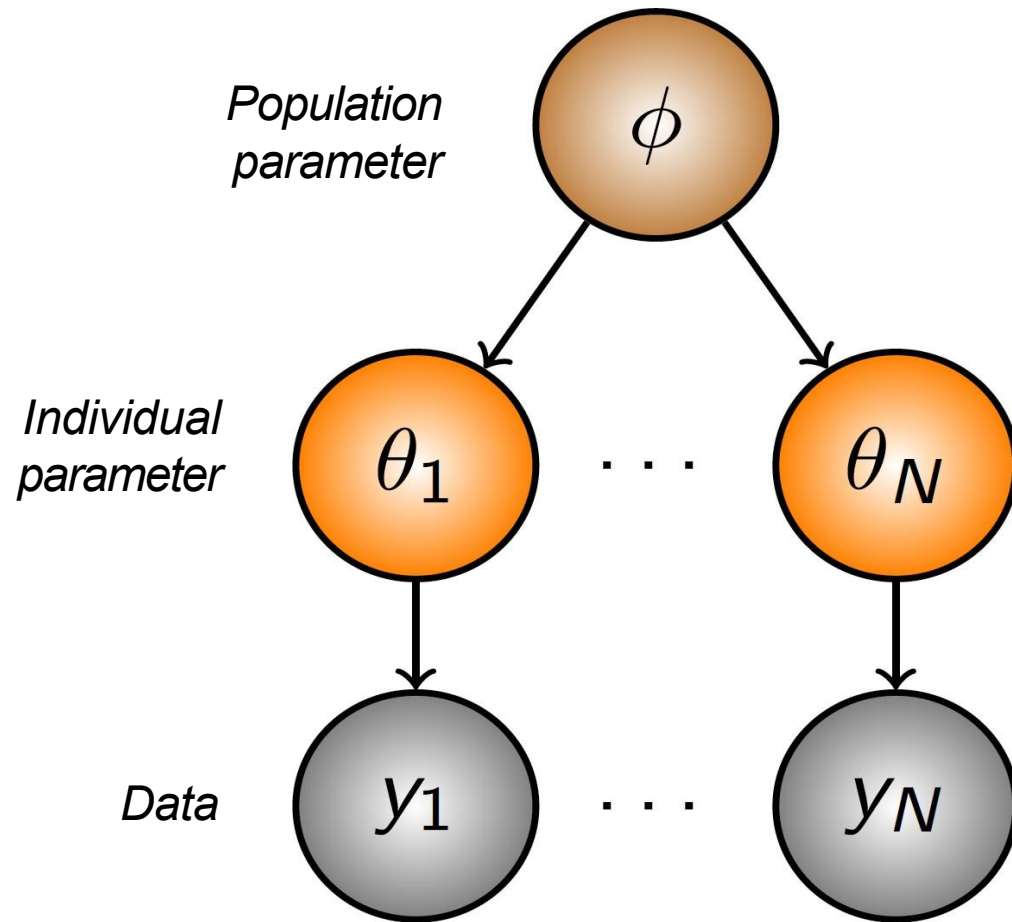
Individual to population-level inferences

Trajectory



Data

Individual to population-level inferences

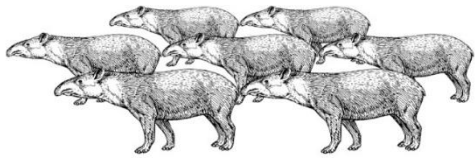


Hierarchical models

- This framework facilitates population-level inference with as few as 2–3 observed *home range crossings* (τ_p) and similarly *small number of individuals* (m).

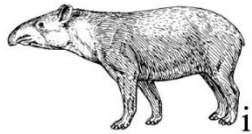
Individual to population-level inferences

Population



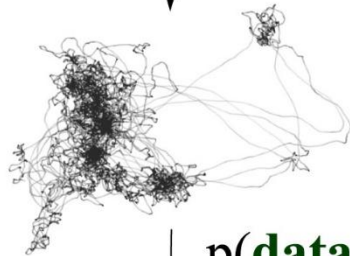
$$\downarrow p(\boldsymbol{\theta}_i | \boldsymbol{\Theta})$$

Individual



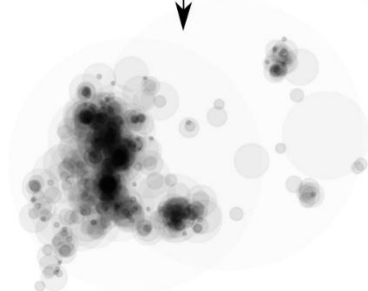
$$\downarrow p(\mathbf{r}_i | \boldsymbol{\theta}_i)$$

Trajectory



$$\downarrow p(\mathbf{data}_i | \mathbf{r}_i)$$

Data



$$p(\mathbf{DATA} | \boldsymbol{\Theta}) = \prod_i^m \int d\mathbf{r}_i \int \boldsymbol{\theta}_i \underbrace{p(\mathbf{data}_i | \mathbf{r}_i)}_{\text{observation}} \underbrace{p(\mathbf{r}_i | \boldsymbol{\theta}_i)}_{\text{movement}} \underbrace{p(\boldsymbol{\theta}_i | \boldsymbol{\Theta})}_{\text{population}},$$

DATA = tracking data from m individuals,

data_i = tracking data from individual i ,

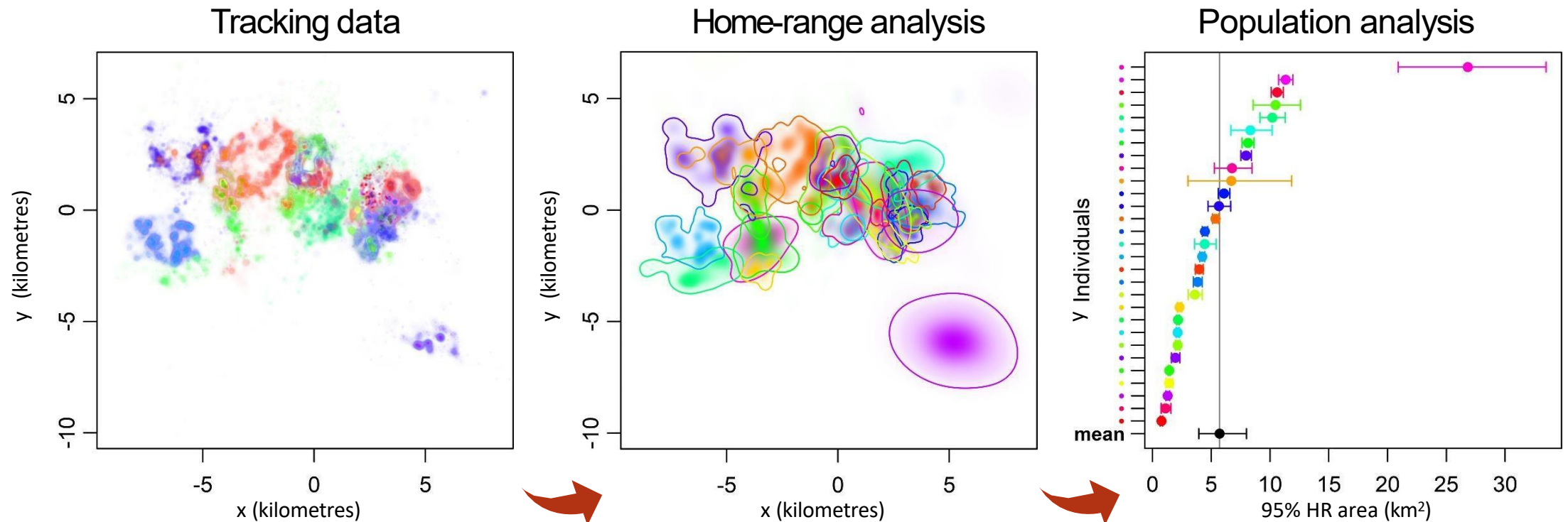
r_i = trajectory of individual i ,

θ_i = movement characteristics (e.g., home range) of individual i ,

Θ = Population parameters (e.g., variation in home ranges).

Individual to population-level inferences

A **statistically efficient estimator** will *downweigh* uncertain estimates relative to more certain estimates, such that the estimated mean has a smaller variance.



Home range area estimates follow
a χ^2 sampling distribution

Population of home-range areas follows
an **inverse-Gaussian distribution**

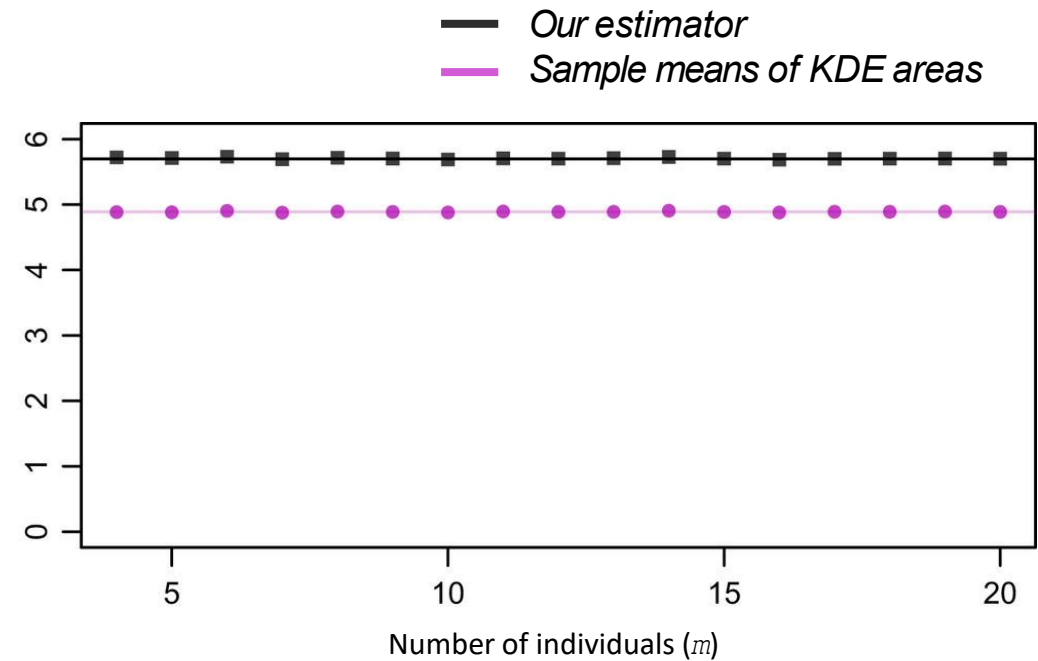
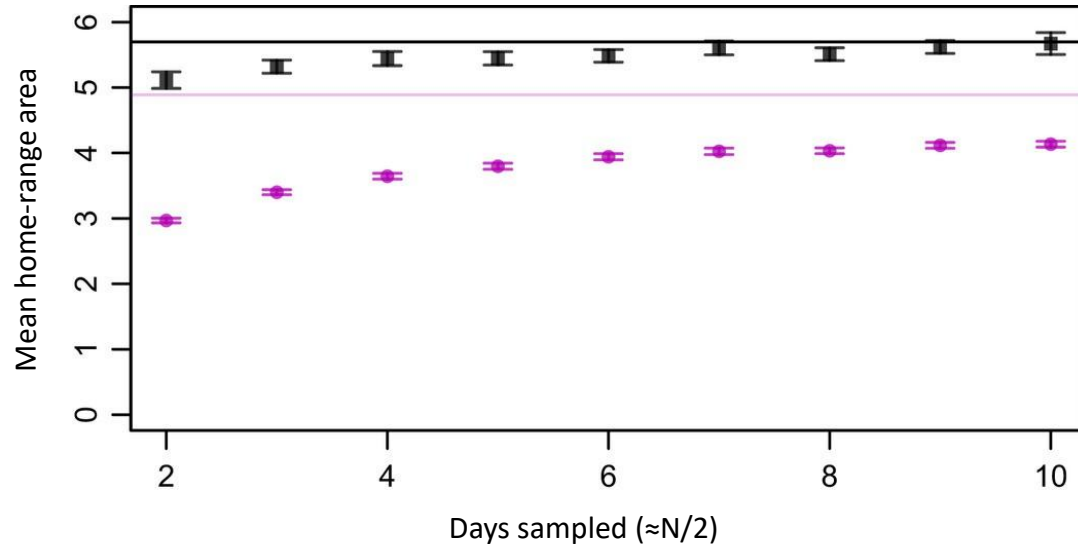
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Applications

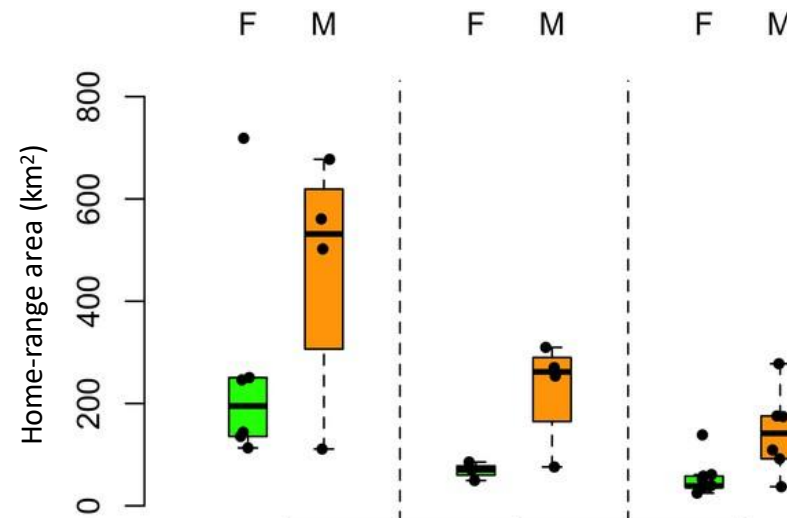


Tapirs have **home range crossing times** (τ_p) of **0.72 days** (ranging from **0.05–12.8 days**)



Applications

Morato et al. (2016)
DOI: 10.1093/jmammal/gyab068



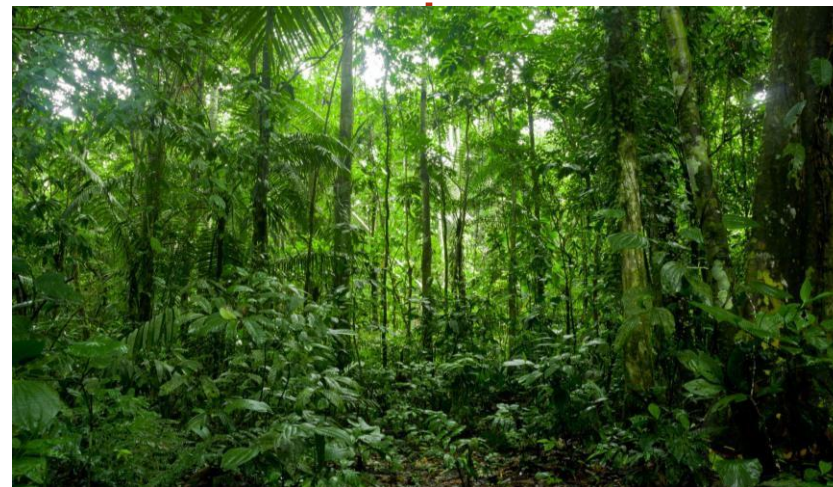
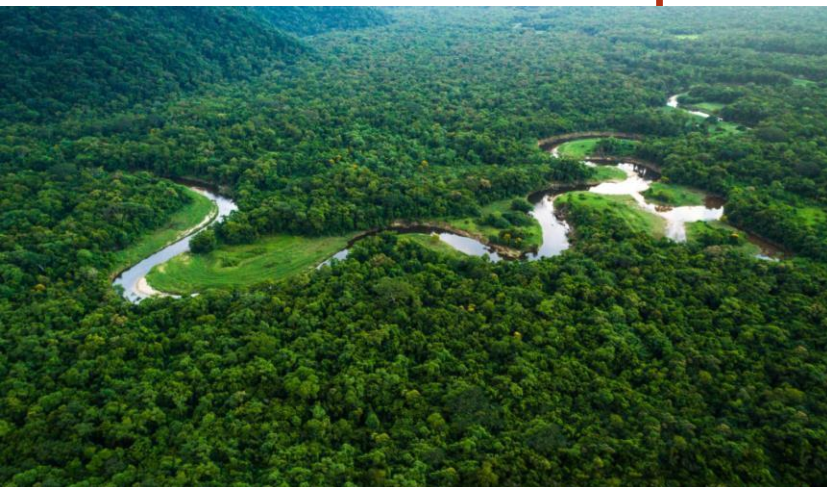
Atlantic forest

Amazon

Pantanal



Pete Wilcox



Frans Lanting



ctmm_akde.R

Thank you

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